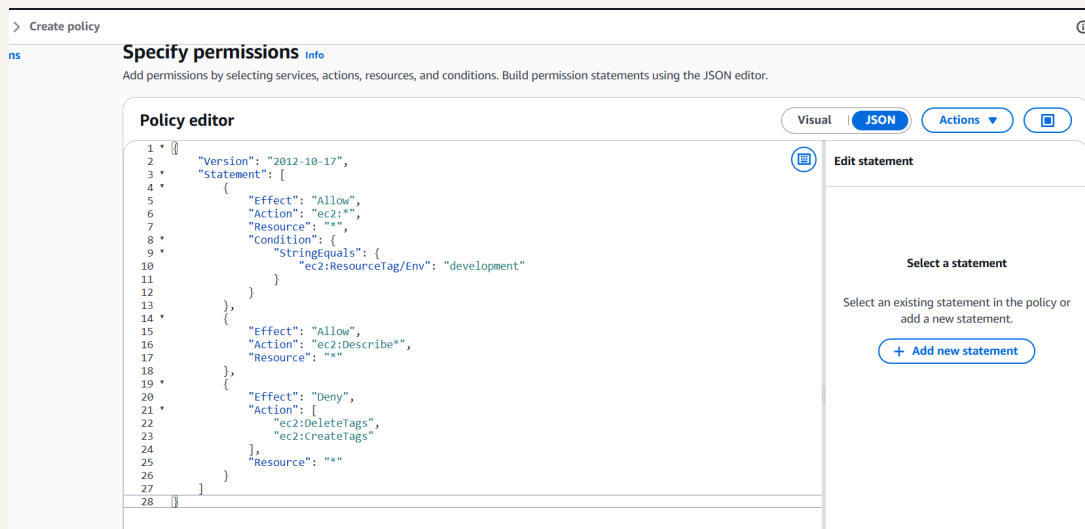




Cloud Security with AWS IAM

D

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Introducing Today's Project!

Project overview

In this project, I will demonstrate my understanding of Cloud Security with AWS IAM (Identity and Access Management)... I'm doing this project to learn and get hands on experience using AWS Cloud Infrastructure.

Tools and concepts

In this project I learned how IAM works to control access to AWS resources. I created a policy that defined exactly what our intern could and couldn't do. I allowed them full access to the development environment while blocking any access to the production environment. I learned how to attach policies to user groups so when users are added to the group, they automatically inherit the right permissions. The biggest takeaway was the principle of least privilege. Only giving users the minimum access they need to do their job.

Project reflection

This project took me approximately 2 hours. The most challenging part was when I tested the IAM policy in the intern's account and deleted the dev instance instead of stopping it. I then reviewed and made changes to the policy, reassigned it to the iam group, logged into the intern's account and tried again. It was most rewarding to know that I understood why it "broke" and how to fix the issue.



Tags

What I did in this step

In this step, I will launch a few EC2 instances so that we are able to boost NextWork's computing power due to the increased traffic to the website over this holiday period.

Understanding tags

Tags are labels that we attach to AWS resources for organisation, structured as key-value pairs. The example we used in this task was a tag with the key "ENV" and values of "dev" and "prod" to distinguish between development and production environments. Tags make it easy to identify, filter, and group related resources. They're also useful for cost tracking because you can break down your AWS bill by tagged projects or teams. And, for access control, where IAM policies can use tags to manage permissions.

My tag configuration

My two Ec2 instances each have two tags. The first instance has a Name tag with the value "nextwork-dev-Demayne" and an Env tag with the value "development". The second instance has a Name tag with the value "nextwork-prod-Demayne" and an Env tag with the value "production". These tags help distinguish which instance is being used for development versus production.



i-0b385d84b311430ef

23.20.86.51 [Public IP]

IAM role

-

IMDSv2

Required

Operator

-

vpc-0ce89e2934bb6614c

Subnet ID

subnet-0f193dcf8e3506eb2

Instance ARN

arn:aws:ec2:us-east-1:479887546653:instance/i-0b385d84b311430ef

Opt-in to AWS Compute Optimizer for recommendation s.

Learn more

Auto Scaling Group name

-

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Tags

Q

Manage tags

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Key

Value

Name

nextwork-prod-Demayne

Env

production



IAM Policies

What I did in this step

In this step, we are IAM (Identity and Access Management) to create a policy that grants our new intern access to only the development EC2 instance. This follows the principle of least privilege giving users the minimum permissions they need to do their job. By restricting the intern to the dev environment, we protect the production instance from accidental shutdowns or untested changes being pushed to live users.

Understanding IAM policies

IAM Policies are rules that define who can do what with AWS resources. They are attached to IAM users, groups or roles, specifying which actions are allowed, on which resources, and under what conditions. They are stored as JSON documents that AWS reads and decides whether to approve or block a request.

The policy I set up

For this project, I've set up a policy using JSON.

Policy effect

I've created a policy that has three effects. First, it allows the intern to perform all EC2 actions on any resource tagged `END = development`. This gives them full control over the dev instance.



Second, it allows the intern to describe all EC2 instances regardless of their tag, so they can view and identify resources accross the account. Third, it explicitly denies the intern the ability to create or delete tags on any instance. Since a Deny always overrides an Allow in IAM, the intern cannot manage tags even on the dev instance despite having access to it.

Understanding Effect, Action, and Resource

Effect can have two values - Allow or Deny. This indicates whether a policy allows or denies a certain action. Deny ALWAYS has priority over Allow. Actions are a list of specific AWS actions that the effect allows or denies. Resource specifies to which AWS resources the policy applies to. A value of "*" means that the policy applies to all resources.



My JSON Policy

Create policy

ns

Specify permissions

Info

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

Visual

JSON

Actions

Policy editor

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```
"Version": "2012-10-17",
"Statement": [
  {
    "Effect": "Allow",
    "Action": "ec2:*",
    "Resource": "*",
    "Condition": {
      "StringEquals": {
        "ec2:ResourceTag/Env": "development"
      }
    }
  },
  {
    "Effect": "Allow",
    "Action": "ec2:Describe*",
    "Resource": "*"
  },
  {
    "Effect": "Deny",
    "Action": [
      "ec2:DeleteTags",
      "ec2:CreateTags"
    ],
    "Resource": "*"
  }
]
```

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

+ Add new statement



Account Alias

What I did in this step

In this step, I will create an AWS account alias in order to simplify user login.

Understanding account aliases

An account alias is a custom, friendly name used for your AWS account that replaces your 12-digit account ID. Instead of signing in with a hard-to-remember numeric URL, you can use your alias. Each AWS account can only have one alias at a time.

Setting up my account alias

Creating an account alias took me less than 60 seconds. Now, my new AWS console sign-in URL is `nextwork-alias-project-intern`.



✓ Alias nextwork-alias-project-intern created for this account.

IAM Dashboard [info](#)

Security recommendations 0

✓ Root user has MFA
Having multi-factor authentication (MFA) for the root user improves security for this account.

✓ You have MFA
Having multi-factor authentication (MFA) for the IAM user improves security for this account.

✓ Your user, Demayne-IAM-Admin, does not have any active access keys that have been unused for more than a year.
Deactivating or deleting unused access keys improves security.

AWS Account

Account ID
 479887546653

Account Alias
nextwork-alias-project-intern [Edit](#) | [Delete](#)

Sign-in URL for IAM users in this account
 <https://nextwork-alias-project-intern.signin.aws.amazon.com/console>



IAM Users and User Groups

What I did in this step

In this step, I will create an IAM user group for all Nextwork interns so I can manage all their permissions in one place. Also, I will create a dedicated IAM user for the new intern so that they have a way to log in.

Understanding user groups

IAM user groups are collections of IAM users grouped together for easier user management.

Attaching policies to user groups

Attaching the policy to the user group means that every member in the group automatically inherits its permissions. As it stands, our intern has full control over the dev EC2 instance, can describe all instances but is blocked from creating or deleting tags. If we onboard more interns, we just add them to the group. No extra configuration needed.

Understanding IAM users

IAM users are identities created within an AWS account that represent people or applications needing access to AWS resources. Each IAM user has their own credentials (a username and password for console access or access keys). Permissions are then assigned to each user, either directly through policies or by adding them to a user group.



Logging in as an IAM User

Sharing sign-in details

The first way is via a downloadable CSV(Comma Separates Value) file containing their console sign-in URL, password and username. The second is to send the user sign-in instructions directly via email from the IAM console during the user creation process.

Observations from the IAM user dashboard

Once I logged in as my IAM user, I noticed a red border around the Applications Tab that read:Access denied to servicecatalog:ListApplications.This was because the IAM policy did not give the interns account access to Applications.

✔ User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

View user

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Step 4
Retrieve password

Retrieve password

You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.

Console sign-in details

Email sign-in instructions ↗

Console sign-in URL
https://nextwork-alias-project-intern.signin.aws.amazon.com/console

User name
nextwork-dev-Demayne

Console password
***** Show

Cancel

Download .csv file

Return to users list



Testing IAM Policies

What I did in this step

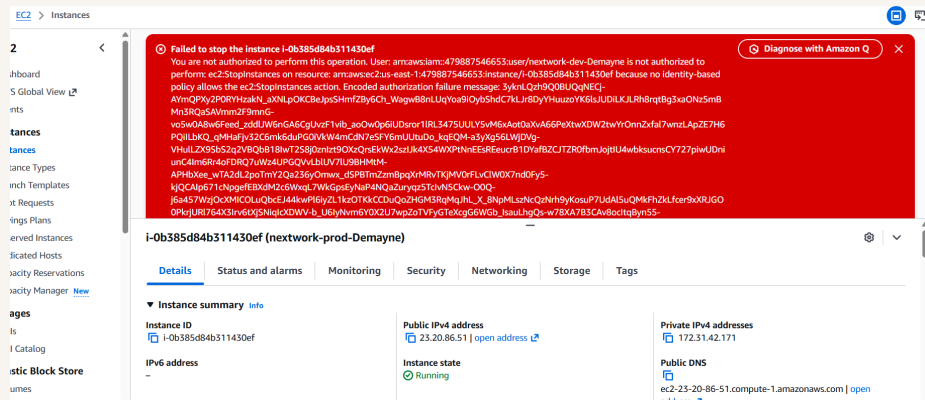
In this step, I will log into the new interns Account so I can test their access to the development and production instances.

Testing policy actions

I tested my JSON IAM policy by checking whether or not the interns account was able to initiate stop states to development and production EC2 instances after I had set up an IAM policy giving the intern access to development environment and not the production environment.

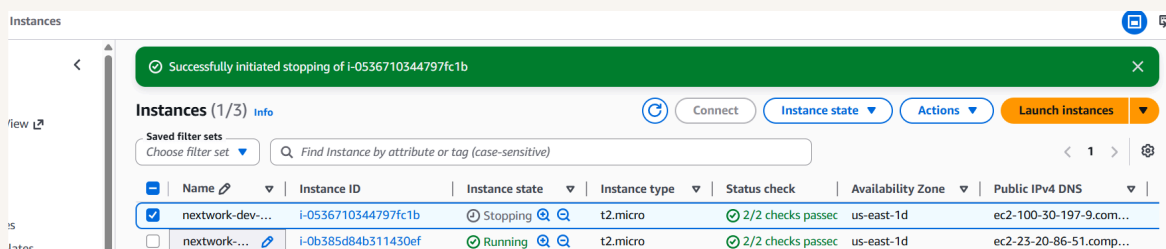
Stopping the production instance

When I tried to stop the production instance, a "failed to stop" red banner appeared on the screen. This was because the intern account is not authorised to stop the production instance.



Stopping the development instance

Next, when I tried to stop the development instance, it stopped successfully. This was because the IAM policy authorised this action for the interns AWS User account.





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